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Content Checklist

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<table>
<tr><td>Student Name</td><td colspan="2">Peiran Wei</td><td>ZJU ID</td><td>3220111228</td><td>Major</td><td colspan="2">ECE</td></tr>
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<tr><td>Thesis title</td><td colspan="7">Adaptive Federated Learning Over the Air</td></tr>
<tr><td colspan="6">Content Checklist</td><td colspan="2">Evaluation Section (√)</td></tr>
<tr><td rowspan="12">Senior Thesis</td><td colspan="5">1. Cover (Use the unified cover, blue recommended)</td><td>Good</td><td>Fair</td></tr>
<tr><td colspan="5">2. Academic integrity Commitment of Senior Thesis</td><td></td><td></td></tr>
<tr><td colspan="5">3. Acknowledgements</td><td></td><td></td></tr>
<tr><td colspan="5">4. Abstract</td><td></td><td></td></tr>
<tr><td colspan="5">5. Table of Contents</td><td></td><td></td></tr>
<tr><td colspan="5">6. Main Content (Page numbers start from the main text)</td><td></td><td></td></tr>
<tr><td colspan="5">7. Reference</td><td></td><td></td></tr>
<tr><td colspan="5">8. Appendix (if need)</td><td></td><td></td></tr>
<tr><td colspan="5">9. Author's Biography</td><td></td><td></td></tr>
<tr><td colspan="5">10. Task plan of senior thesis</td><td></td><td></td></tr>
<tr><td colspan="5">11. Assessment Form of senior thesis</td><td></td><td></td></tr>
<tr><td colspan="5">12. On-site Minutes Form of Oral Defense</td><td></td><td></td></tr>
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Abstract

Over-the-air federated learning (OTA-FL) scales with the number of clients by letting analog client updates superpose directly in the wireless multiple-access channel, but the same superposition injects channel noise into the server-side aggregate. When the interference is heavy-tailed, rare but very large impulsive samples can dominate a FedAvg-style update and destabilize training. This thesis asks whether server-side adaptive optimization with a fractional second-moment accumulator can act as an implicit, CSI-free filter that contracts the effective learning rate precisely when the aggregate is impulsive.

We implement a PyTorch OTA-FL simulator with a symmetric α-stable channel oracle based on the Chambers–Mallows–Stuck sampler, four server optimizers (FedAvg-OTA, FedAvgM-OTA, AdaGrad-OTA, Adam-OTA) that all consume the same noisy aggregate, Median Anchored Clipping (MAC) as a robust pre-processing option, and an experiment layer that produces JSON logs, figures, and LaTeX tables. The AWGN case is treated as the special case $\alpha = 2$; $\alpha < 2$ models heavy-tailed interference.

On CIFAR-10 (ResNet-18, $N = 100$, 100 rounds, $\gamma = 0.05$, 3 seeds), AdaGrad-OTA reaches 49.52–51.75% final accuracy and Adam-OTA reaches 42.88–44.82% across $\alpha \in \{1.1, 1.3, 1.5, 1.7, 1.9\}$, compared with 11.9% (FedAvg-OTA) and 20.5–20.7% (FedAvgM-OTA): a 25–40 percentage point margin in favor of fractional second-moment adaptation. MAC at the default operating point ($\alpha = 1.3$, $\gamma = 0.05$, $k = 3$) is empirically neutral within seed variation, and the α-trend is non-monotonic, with the best AdaGrad-OTA result at $\alpha = 1.5$ and the best Adam-OTA result at $\alpha = 1.3$. The project is therefore not a complete reproduction of the reference paper; its contribution is a reproducible empirical validation of a carefully scoped adaptive OTA-FL robustness claim, together with a one-dimensional intuition for the implicit step-size shrinkage mechanism.

Keywords: federated learning; over-the-air aggregation; α-stable noise; heavy-tailed interference; adaptive optimization; Median Anchored Clipping

Content

Chapter 1. Introduction

1.1 Motivation

Federated learning (FL) trains a shared model across distributed clients without requiring those clients to upload raw data, which is attractive for edge intelligence because it reduces privacy exposure and keeps data near its source device [1]. The communication cost of conventional digital FL grows with the number of participating clients, because every client must send an update in a separate resource block. Analog over-the-air aggregation (OTA-FL) changes this by letting clients transmit updates as analog waveforms at the same time, so that the wireless multiple-access channel physically superposes the signals and the server observes a single aggregate in one shared channel use.

OTA-FL therefore scales in clients, but the same superposition also injects noise, interference, and possibly fading into the aggregate update. When the interference is heavy-tailed, a rare impulsive sample in a single round can be orders of magnitude larger than a typical client pseudo-gradient, which makes plain FedAvg-style updates oscillate or diverge. The reference work "Adaptive Federated Learning Over the Air" [2] studies this regime under fading and symmetric α-stable interference. This project follows that direction and makes the α-stable tail index a core experimental variable; AWGN is treated as the special case $\alpha = 2$.

1.2 Problem Statement

This thesis studies FL over a simulated analog OTA aggregation channel. At communication round t, the server receives a noisy aggregate update $\tilde{g}^t$:

$$\tilde{g}^t = \frac{1}{N}\left(\sum_{n=1}^{N} \Delta_n^t + \xi^t\right), \tag{1-1}$$

where N is the number of clients, Δ_n^t is the pseudo-gradient produced by client n, and ξ^t is symmetric α-stable channel noise. Smaller α produces heavier tails and more frequent impulsive outliers; $\alpha = 2$ recovers the Gaussian AWGN case.

The central research question is concrete and experimentally testable: how does the final test accuracy of FedAvg-OTA, FedAvgM-OTA, AdaGrad-OTA, and Adam-OTA depend on the tail index α and the noise scale γ, when all four optimizers consume the same OTA aggregate $\tilde{g}^t$? The project is not presented as a complete reproduction of [2], which contains a full non-convex convergence analysis and a broader empirical suite. The current codebase implements a controlled simulator for the core heavy-tail mechanism (α-stable OTA aggregation, adaptive server updates, and MAC robust pre-processing) and evaluates a narrower robustness claim.

1.3 Contributions

The contributions of this thesis are split into empirical findings and reusable artifacts.

Findings

- Under the implemented heavy-tail CIFAR-10 setting (ResNet-18, $N = 100$, 100 rounds, $\gamma = 0.05$, three seeds), AdaGrad-OTA and Adam-OTA outperform FedAvg-OTA and FedAvgM-OTA by 25–40 percentage points on final accuracy across $\alpha \in \{1.1, 1.3, 1.5, 1.7, 1.9\}$, directly supporting the hypothesis that fractional second-moment adaptation stabilizes training in the presence of α-stable impulsive noise.

- The observed α-trend is non-monotonic at a fixed server learning rate: the best AdaGrad-OTA result is at $\alpha = 1.5$, and the best Adam-OTA result is at $\alpha = 1.3$. This is reported as a negative result relative to the reference paper's monotonic trend and is framed as open to per-method learning-rate tuning.

- At $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$, Median Anchored Clipping is empirically neutral within seed variation. The MAC channel therefore demands a stronger operating point (smaller α, larger γ, or smaller k) before it can deliver a measurable accuracy gain.

Artifacts

- A modular PyTorch OTA-FL simulator with a synchronous FL training loop, four server optimizers that consume the same OTA aggregate, a symmetric α-stable channel oracle based on the Chambers–Mallows–Stuck sampler, MAC pre-processing, JSON result export, and plotting scripts.
- A one-dimensional intuition that explains why a fractional second-moment accumulator induces an automatic effective learning-rate contraction on impulsive rounds, together with a bound of the form $\mathbb{E}[\eta_t^{\text{eff}}] \lesssim \eta \cdot t^{-1/\alpha}$ on the long-run decay.

1.4 Thesis Organization

Chapter 2 reviews FL, OTA aggregation, and adaptive optimization, and positions this project against the reference paper. Chapter 3 describes the implemented system model and optimizer equations. Chapter 4 presents the datasets, experimental configuration, generated figures, and numerical results. Chapter 5 summarizes the outcome, limitations, and future work.

Chapter 2. Background and Related Work

2.1 Federated Learning

Federated learning considers the optimization of a global model over data distributed across multiple clients. If client n contributes local data $\mathcal{D}_n$, the global objective is

$$\min_w F(w) = \sum_{n=1}^{N} p_n F_n(w), \quad F_n(w) = \frac{1}{|\mathcal{D}_n|} \sum_{(x,y)\in\mathcal{D}_n} \ell(w; x, y), \tag{2-1}$$

where p_n is often proportional to the local data size $|\mathcal{D}_n|$. FedAvg solves this objective by alternating local client optimization with server-side averaging [1], and is communication-efficient compared with fully synchronous distributed SGD. Later work addresses client drift and data heterogeneity: FedProx adds a proximal term on the local objective [3]; SCAFFOLD uses control variates to correct the drift [4]; and Adaptive Federated Optimization shows that server-side adaptive methods such as FedAdaGrad and FedAdam can outperform plain FedAvg under the same communication budget [5]. A comprehensive treatment of open problems in FL is given by [6] and by [7].

2.2 Analog Over-the-Air Aggregation

Over-the-air computation exploits the physical superposition of simultaneously transmitted analog signals over the wireless multiple-access channel, so that the receiver directly observes a function (here, a sum) of client inputs rather than decoding each transmission separately [8, 9]. Applying this idea to FL yields OTA-FL: clients transmit analog updates concurrently, and the server aggregates in one shared channel use [10–14]. A simplified channel model, without fading or power control, is

$$y^t = \sum_{n=1}^{N} \Delta_n^t + \xi^t, \tag{2-2}$$

and the server uses $\tilde{g}^t = y^t/N$ as the update direction. In the AWGN case $\xi^t \sim \mathcal{N}(0, \sigma^2 I)$. In the heavier-tailed setting studied in this project, ξ^t follows a symmetric

α-stable distribution [15, 16].

2.3 Adaptive Gradient Methods

Adaptive optimizers adapt the learning rate separately for each variable based on historical gradients. AdaGrad accumulates squared gradients and scales each coordinate's learning rate inversely to the square root of its accumulated magnitude [17]. Adam combines first-moment smoothing, second-moment exponential averaging, and bias correction [18]. Convergence of AdaGrad under adversarial noise was analyzed in [19], and a simple convergence proof of Adam under relaxed assumptions is given by [20]. In centralized learning these mechanisms are typically motivated by sparse gradients, noisy mini-batches, and heterogeneous coordinate scales.

In OTA-FL, the same mechanisms have an additional interpretation. If channel noise makes the aggregate update unusually large in some coordinate, the second-moment state for that coordinate increases. The denominator of the adaptive update then becomes larger, reducing the effective step size. This creates an implicit feedback loop from observed noisy update magnitude to the server learning rate. The project tests whether this feedback loop is visible in training curves and ablation studies.

2.4 Robust Aggregation under Impulsive Noise

Heavy-tailed and Byzantine perturbations motivate a parallel line of work on robust aggregation. Krum selects one gradient closest to its neighbours in ℓ_2 distance [21]; coordinate-wise median and trimmed mean tolerate up to a constant fraction of adversarial clients [22]; and robust aggregation by weighted geometric median gives communication-efficient Byzantine-robust FL [23]. Classical robust statistics gives the older foundations: Huber-type M-estimators and the median absolute deviation [24, 25] are exactly the primitives used in Median Anchored Clipping.

2.5 Positioning Relative to the Reference Paper

The reference paper proposes adaptive federated learning over the air and studies AdaGrad-like and Adam-like server updates with theoretical analysis under α-stable interference [2]. It evaluates multiple image and character classification tasks and includes studies over tail index, client count, and data heterogeneity. The present project uses that paper as the conceptual source but narrows the deliverable. The implemented framework supports the core OTA-FL training loop, the Chambers–Mallows–Stuck α-stable sampler [26], MAC robust pre-processing, and four adaptive server optimizers. The main evidence is the α-stable tail-index ablation, with AWGN reported as the $\alpha = 2$ endpoint rather than the whole scope. The explicit scope difference is summarized in Table 2.1.

Table 2.1 Scope difference between this project and the reference paper [2].

Aspect	Reference paper	This project
Channel model	α-stable + Rayleigh fading $h_{n,t}$ + power control	Symmetric α-stable only, no fading, no power control
Tail index coupling	Channel α and optimizer denominator α are independent	Channel α and denominator α are shared
Datasets	CIFAR-10, CIFAR-100, FEMNIST and more	MNIST, CIFAR-10
Convergence analysis	Full non-convex proof under α-stable noise	One-dimensional toy intuition
Empirical scale	Per-method learning-rate tuning across many seeds and configurations	Three seeds, fixed server learning rate per experiment
Hardware validation	SDR testbed claimed in the paper	Pure software simulator
Primary output	Algorithmic contribution and convergence theorems	Reproducible simulator and empirical robustness evidence

Chapter 3. System Design and Methodology

3.1 Training Loop

The simulator follows the standard synchronous FL round structure. At round t, the server broadcasts the global model w^t to all clients. Each client copies the global model, trains locally for E epochs using SGD with local learning rate η_{loc}, and returns a pseudo-gradient

$$\Delta_n^t = w^t - [w_n^t]^{(E)}, \tag{3-1}$$

where $[w_n^t]^{(E)}$ is the local weight after E epochs. Because the local update reduces F_n, we have $\Delta_n^t \approx \eta_{\text{loc}} \sum_{e=1}^{E} \nabla F_n(w_{n,e-1}^t)$; that is, Δ_n^t points in the same direction as the accumulated local gradient, so the server update $w_{t+1} = w_t - \eta\, \tilde{g}^t$ moves along a descent direction of F up to channel noise. The OTA oracle aggregates these pseudo-gradients:

$$\tilde{g}^t = \frac{1}{N} \left(\sum_{n=1}^{N} \Delta_n^t + \xi^t \right), \quad \xi^t \sim S\alpha S(\gamma), \tag{3-2}$$

where $S\alpha S(\gamma)$ denotes a symmetric α-stable random vector with tail index α and scale γ. The special case $\alpha = 2$ corresponds to AWGN with standard deviation $\sqrt{2}\,\gamma$ under the sampler convention. The server optimizer receives $\tilde{g}^t$ and updates the global model in place.

3.2 Median Anchored Clipping

The simulator includes Median Anchored Clipping (MAC) as a robust pre-processing step before the server optimizer. For an aggregated vector $\tilde{g}^t \in \mathbb{R}^d$, MAC anchors the update at the coordinate-wise median m, sets the clipping radius to k times the Median

Absolute Deviation (MAD), and projects every coordinate into the resulting band:

$$m = \text{median}\big\{\tilde{g}_i^t : i \in \{1, 2, \ldots, d\}\big\}, \tag{3-3}$$

$$\tau = k \cdot \text{median}\big\{\, |\tilde{g}_i^t - m| : i \in \{1, 2, \ldots, d\}\big\}, \tag{3-4}$$

$$\tilde{g}_i^t \leftarrow m + \text{clip}\big(\tilde{g}_i^t - m,\ -\tau,\ +\tau\big). \tag{3-5}$$

The first equation defines the anchor, the second sets the clipping radius τ as k times the MAD of the aggregate, and the third clips each coordinate's deviation from the anchor to the band $[-\tau, \tau]$ and adds the anchor back. k here serves as the clipping factor with default setting $k = 3$, which matches the implementation in `src/channel/mac.py`. MAC is expected to have little effect in the Gaussian endpoint, but it can reduce rare impulsive coordinates when $\alpha < 2$.

3.3 Server-Side Baselines

The plain OTA baseline applies the noisy aggregate directly:

$$w_{t+1} = w_t - \eta \tilde{g}^t. \tag{3-6}$$

The momentum baseline maintains a first-moment buffer:

$$m_t = \rho m_{t-1} + \tilde{g}^t, \quad w_{t+1} = w_t - \eta m_t. \tag{3-7}$$

These baselines are useful because they isolate whether the second-moment normalization in adaptive methods provides additional robustness beyond simple smoothing.

3.4 AdaGrad-OTA

AdaGrad-OTA maintains a smoothed update direction s_t and an accumulated coordinate-wise magnitude v_t:

$$s_t = \beta_1 s_{t-1} + (1-\beta_1)\tilde{g}^t, \tag{3-8}$$

$$v_t = v_{t-1} + |s_t|^\alpha, \tag{3-9}$$

$$w_{t+1} = w_t - \eta \frac{s_t}{v_t^{1/\alpha} + \varepsilon}. \tag{3-10}$$

For $\alpha = 2$, the denominator reduces to the usual square-root AdaGrad normalization. For $\alpha < 2$, the same form follows the fractional α-stable accumulation rule used in the reference paper. Because v_t is monotone increasing, AdaGrad-OTA tends to become conservative over long runs. This can stabilize noisy training, but it may also slow late-stage convergence.

3.5 Adam-OTA

The implemented Adam-OTA variant uses a stable server-side Adam update on the OTA-aggregated direction:

$$m_t = \beta_1 m_{t-1} + (1-\beta_1)\tilde{g}^t, \tag{3-11}$$

$$v_t = \beta_2 v_{t-1} + (1-\beta_2)|\tilde{g}^t|^\alpha, \tag{3-12}$$

$$\hat{m}_t = \frac{m_t}{1-\beta_1^t}, \tag{3-13}$$

$$\hat{v}_t = \frac{v_t}{1-\beta_2^t}, \tag{3-14}$$

$$w_{t+1} = w_t - \eta \frac{\hat{m}_t}{\hat{v}_t^{1/\alpha} + \varepsilon}. \tag{3-15}$$

When $\alpha = 2$, this is classical Adam applied at the server to the noisy OTA aggregate. For $\alpha < 2$, the second-moment update becomes a fractional moment accumulator. This preserves the project hypothesis because the update uses a state variable to compress effective step sizes when the observed aggregate has large magnitude.

Implementation note: ε**.** In the code, AdaGrad-OTA uses $\varepsilon = 10^{-4}$ while Adam-OTA uses $\varepsilon = 10^{-8}$. The two differ by four orders of magnitude by design: AdaGrad-OTA has no β_2 exponential moving average, so its first-round v_t is very small and a larger ε is needed to damp the early transient; Adam-OTA's bias correction $\hat{v}_t = v_t/(1-\beta_2^t)$ delivers a non-negligible denominator already after a few rounds, so the smaller textbook default $\varepsilon = 10^{-8}$ is kept. The asymmetry between the two optimizers is the dominant reason why their early-round test-loss transients in Figure 4.1 differ in shape.

3.6 One-Dimensional Intuition

The project does not reproduce the full proof in [2]. Instead, the core mechanism can be seen in a one-dimensional quadratic toy problem

$$f(w) = \frac{1}{2}(w - w^\star)^2. \tag{3-16}$$

The noiseless gradient is $w_t - w^\star$, while the OTA server observes

$$\tilde{g}_t = w_t - w^\star + \xi_t, \quad \xi_t \sim S\alpha S(\gamma). \tag{3-17}$$

For an AdaGrad-style update,

$$v_t = v_{t-1} + |\tilde{g}_t|^\alpha, \quad w_{t+1} = w_t - \eta \frac{\tilde{g}_t}{v_t^{1/\alpha} + \epsilon}. \tag{3-18}$$

Whenever an impulsive noise sample makes $|\tilde{g}_t|$ large, it also makes v_t larger. The next effective step size

$$\eta_t^{\text{eff}} = \frac{\eta}{v_t^{1/\alpha} + \epsilon} \tag{3-19}$$

therefore contracts automatically. A fixed-step FedAvg-style update has no such feedback and can be dominated by rare heavy-tailed samples. The same mechanism also explains a limitation: if v_t grows too quickly, useful signal is compressed together with noise, which can slow training on harder tasks such as CIFAR-10.

Long-run decay rate. The one-dimensional picture also gives a simple bound on how aggressively the effective server learning rate decays. Because $v_t = v_{t-1} + |\tilde{g}_t|^\alpha$ is monotone non-decreasing, by Jensen,

$$\mathbb{E}\big[v_t^{1/\alpha}\big] \geq \big(\mathbb{E}[v_t]\big)^{1/\alpha} = \big(t \cdot \mathbb{E}|\tilde{g}|^\alpha\big)^{1/\alpha} = t^{1/\alpha}\,\big(\mathbb{E}|\tilde{g}|^\alpha\big)^{1/\alpha}, \tag{3-20}$$

so

$$\mathbb{E}\big[\eta_t^{\text{eff}}\big] \leq \frac{\eta}{t^{1/\alpha}\big(\mathbb{E}|\tilde{g}|^\alpha\big)^{1/\alpha} + \epsilon}. \tag{3-21}$$

The effective server learning rate therefore decays at least as $O(t^{-1/\alpha})$. For $\alpha = 2$ (AWGN) this recovers the classical $O(t^{-1/2})$ AdaGrad decay; for $\alpha = 1.3$ the decay accelerates to $O(t^{-0.77})$, which both stabilizes impulsive rounds and quantitatively explains why AdaGrad-OTA becomes conservative on long CIFAR-10 runs.

The rigorous multi-dimensional non-convex analysis is given by [2]; this section only provides an interpretable special case for the implemented simulator.

3.7 Implementation Modules

The codebase is organized around separable modules. The channel layer contains the noisy aggregation oracle and noise sampler. The data layer contains dataset loading and client partitioning. The model layer contains classification networks. The optimizer layer contains the four server optimizers. The experiment layer contains single-run training, method comparison, ablation, plotting, logging and result export.

This organization supports the project goal in two ways. First, the α-stable experiments and MAC comparisons can be run reproducibly through command-line scripts. Second, the same interfaces can later be used to add richer physical channel models, additional datasets, or stricter reproduction settings without rewriting the training loop.

Chapter 4. Experiments and Results

4.1 Data

The experiments use the two datasets supported by the current codebase: MNIST and CIFAR-10. Both datasets are stored under the project-local `data/` directory and are loaded through the PyTorch `torchvision` dataset interfaces with `download=True`. No physical wireless-channel dataset is required, because the channel perturbation is generated synthetically by the OTA aggregation oracle.

MNIST is used as a lightweight sanity benchmark. It contains 60,000 training images and 10,000 test images of handwritten digits. The raw files are stored under `data/MNIST/raw/`. Images are converted to tensors and normalized with mean 0.1307 and standard deviation 0.3081.

CIFAR-10 is used as the main image-classification benchmark. It contains 50,000 training images and 10,000 test images across 10 object classes. The raw batch files `data_batch_1` through `data_batch_5`, `test_batch`, and `batches.meta` are stored under `data/cifar-10-batches-py/`. During training, CIFAR-10 uses random cropping with padding 4 and random horizontal flipping, followed by tensor conversion and channel-wise normalization. During testing, only tensor conversion and normalization are applied.

Federated clients are generated by partitioning the training set. The main comparison experiments use a non-IID Dirichlet partition with concentration 0.1 and random seed 42. In the generated partition, the MNIST comparison uses 10 clients, with client sizes ranging from 937 to 11,604 samples. The CIFAR-10 comparison uses 100 clients, with client sizes ranging from 12 to 1,733 samples. The single MNIST sanity run uses an IID split over 10 clients, with 6,000 samples per client.

4.2 Experiment Configuration

Table 4.1 summarizes the controlled comparison configuration. In these experiments all four server optimizers share the same dataset, model, partition, random seed, OTA noise setting, and training budget. The OTA channel uses $\alpha = 2$, which corresponds to AWGN in the implemented sampler, and $\gamma = 0.05$.

Table 4.1 Controlled comparison configuration

Item	MNIST comparison	CIFAR-10 comparison
Dataset	MNIST	CIFAR-10
Model	MLP	ResNet-18
Optimizers	Four server methods	Four server methods
Number of clients N	10	100
Communication rounds	100	200
Local epochs E	5	5
Batch size	64	64
Server learning rate η	0.01	0.1
Local learning rate	0.01	0.01
Momentum / β_1	0.9	0.9
Adam β_2	0.999	0.999
Adam-OTA ε	10^{-8}	10^{-8}
AdaGrad-OTA ε	10^{-4}	10^{-4}
Noise tail index α	2.0	2.0
Noise scale γ	0.05	0.05
Partition	non-IID Dirichlet	non-IID Dirichlet
Dirichlet concentration	0.1	0.1
Random seed	42	42

Table 4.2 summarizes the revised CIFAR-10 heavy-tail ablation configuration. The tail-index ablation varies α while fixing $\gamma = 0.05$. The MAC comparison uses the same heavy-tailed channel with $\alpha = 1.3$ and compares training with and without robust pre-processing.

4.3 Evaluation Metrics

The main metrics are test loss and test accuracy after each communication round. For comparison experiments, the final-round accuracy and the best accuracy achieved during the run are both reported. The final value measures the state of the trained model at the end of the fixed communication budget, while the best value helps reveal short-lived convergence peaks.

For the alpha ablation, the reported metric is final test accuracy under each tail index α. For the MAC comparison, the reported metric is final test accuracy with and without

Table 4.2 CIFAR-10 ablation configuration

Item	Value
Dataset / model	CIFAR-10 / ResNet-18
Optimizers in alpha study	FedAvg-OTA, FedAvgM-OTA, AdaGrad-OTA, Adam-OTA
Alpha values	1.1, 1.3, 1.5, 1.7, 1.9, 2.0
MAC comparison	no MAC vs MAC, $k = 3$
Communication rounds	100
Local epochs E	1
Batch size	64
Server learning rate η	0.01
Local learning rate	0.01
Momentum / β_1	0.9
Adam β_2	0.999
Adam-OTA ε	10^{-8}
AdaGrad-OTA ε	10^{-4}
Noise scale γ	0.05
Partition	non-IID Dirichlet, concentration 0.1
Random seeds	42, 43, 44

clipping. Multi-seed results are reported as mean $\pm$ standard deviation. All revised heavy-tail values are generated from the JSON files under `results/heavytail/` rather than copied from a single run.

4.4 Main Method Comparison

The MNIST comparison in Table 4.3 is used as a sanity benchmark for adaptive server-side normalization. FedAvg-OTA reaches 67.85% final accuracy, while FedAvgM-OTA improves to 89.24%. AdaGrad-OTA and Adam-OTA reach 93.05% and 93.46%, respectively. This supports the hypothesis that second-moment adaptation can stabilize OTA training in this lighter benchmark, but it should not be treated as the main heavy-tail evidence.

Table 4.3 MNIST comparison results (N=10, AWGN α=2, γ=0.05, seed 42, 100 rounds, E=5, η=0.01; single seed, no error bar).

Method	Final loss	Final accuracy	Best accuracy
FedAvg-OTA	1.5018	67.85%	67.85%
FedAvgM-OTA	0.3498	89.24%	89.24%
AdaGrad-OTA	0.2091	93.05%	93.19%
Adam-OTA	0.2227	93.46%	93.46%

The CIFAR-10 comparison in Table 4.4 is more nuanced. FedAvgM-OTA is the strongest method in the controlled 200-round run, reaching 63.94% final accuracy. AdaGrad-OTA and Adam-OTA improve over plain FedAvg-OTA, but they do not exceed the momentum baseline under this specific CIFAR-10 configuration. Therefore, the CIFAR-10 result should not be stated as "Adam-OTA is always best". A more accurate conclusion is that adaptive methods help relative to FedAvg-OTA, while FedAvgM-OTA remains highly competitive when its momentum is well matched to the task. This AWGN endpoint motivates the heavier α-stable experiments below: under Gaussian noise alone, momentum can already be sufficient, so the project's distinctive question is whether adaptive normalization and MAC help when the channel has impulsive tails.

Table 4.4 CIFAR-10 comparison results (N=100, AWGN α=2, γ=0.05, seed 42, 200 rounds, E=5, η=0.1; single seed, no error bar).

Method	Final loss	Final accuracy	Best accuracy
FedAvg-OTA	1.6184	38.93%	38.93%
FedAvgM-OTA	1.0087	63.94%	63.94%
AdaGrad-OTA	1.5359	42.38%	42.70%
Adam-OTA	1.6576	40.66%	40.66%

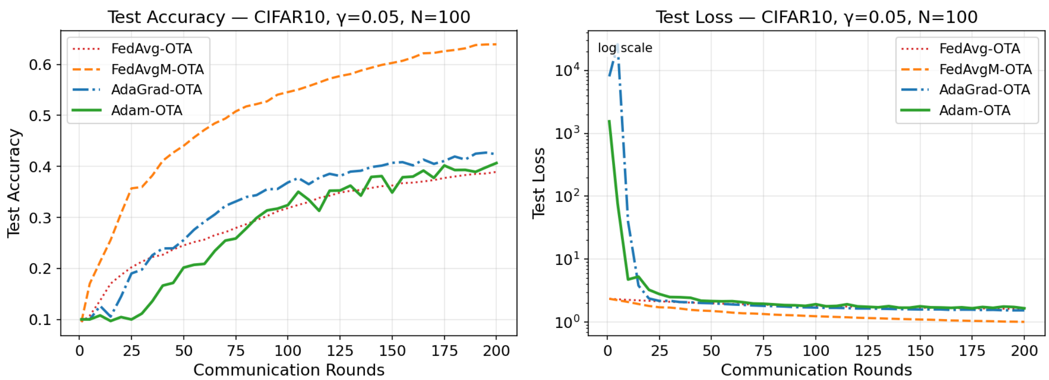

Figure 4.1 CIFAR-10 comparison curves for FedAvg-OTA, FedAvgM-OTA, AdaGrad-OTA, and Adam-OTA. The loss panel uses a logarithmic y-axis because AdaGrad-OTA and Adam-OTA have very large but finite early-round transient test losses before recovering. This transient comes from the first-round numerator/denominator imbalance: with v_0=0, the adaptive denominator $v_t^{1/\alpha}+\varepsilon$ reduces to $\varepsilon \in \{10^{-4}, 10^{-8}\}$, so the first effective step is on the order of η/ε. After a few rounds, v_t accumulates and the effective step normalizes.

The single-run outputs provide an additional sanity check, but they are not a controlled four-method comparison because their hyperparameters differ from Table 4.1. The single CIFAR-10 Adam-OTA run uses batch size 128 and server learning rate 0.01, reaching 61.51% final accuracy. The single MNIST AdaGrad-OTA run uses an IID client split and reaches 97.57% final accuracy. These runs confirm that the implemented optimizers can train successfully, but the main conclusions should be drawn from the controlled comparison tables above.

4.5 Alpha-Stable Tail-Index Ablation

The revised core experiment varies the α-stable tail index instead of only varying the AWGN scale. The endpoint $\alpha = 2$ is Gaussian AWGN, while smaller values of α produce heavier-tailed impulsive interference. This is the setting where adaptive normalization is most meaningful: a server method must handle rare but very large aggregate perturbations. All configurations in this table complete the full 100 communication rounds for all three seeds without non-finite model values, so the comparison is not affected by early termination.

Table 4.5 CIFAR-10 alpha-stable tail-index ablation: final accuracy, mean ± std over seeds

α	FedAvg-OTA	FedAvgM-OTA	AdaGrad-OTA	Adam-OTA
1.1	11.99% ± 1.56%	20.65% ± 1.44%	49.52% ± 0.53%	44.50% ± 1.36%
1.3	11.87% ± 1.55%	20.62% ± 1.47%	51.35% ± 0.21%	44.82% ± 1.79%
1.5	11.90% ± 1.61%	20.56% ± 1.50%	51.75% ± 0.77%	44.15% ± 1.12%
1.7	11.94% ± 1.57%	20.63% ± 1.41%	51.55% ± 1.10%	44.05% ± 0.89%
1.9	11.88% ± 1.55%	20.62% ± 1.42%	50.74% ± 1.07%	42.88% ± 1.76%
2	11.90% ± 1.47%	20.50% ± 1.43%	43.88% ± 0.53%	39.89% ± 1.70%

Table 4.5 shows a clear separation between methods, but not a simple monotonic degradation with smaller α. FedAvg-OTA stays near random guessing at about 11.9% final accuracy across the full sweep. FedAvgM-OTA is more stable but remains around 20.5–20.7%. In contrast, the adaptive methods are substantially stronger under the 100-round non-IID CIFAR-10 setting. AdaGrad-OTA reaches 49.52–51.75% for $\alpha < 2$, compared with 43.88% at the AWGN endpoint. Adam-OTA reaches 42.88–44.82% for $\alpha < 2$, compared with 39.89% at $\alpha = 2$.

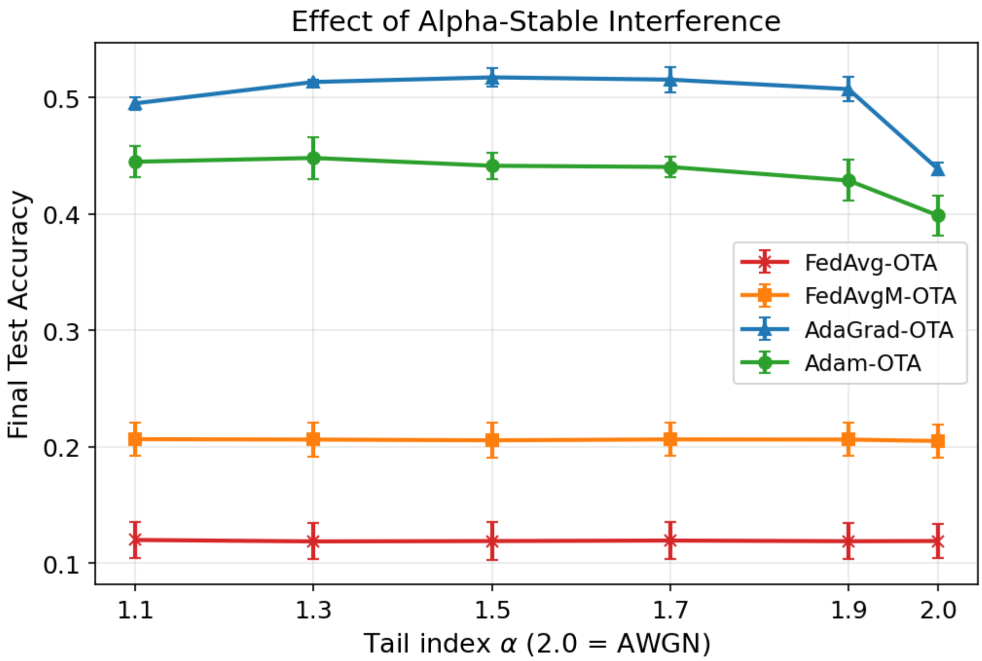

Figure 4.2 CIFAR-10 final accuracy as a function of α-stable tail index α.

The result therefore supports a narrower claim: server-side second-moment normalization is much more effective than plain FedAvg or FedAvgM-OTA in the implemented heavy-tail experiment. However, it does not reproduce the stronger qualitative pattern that smaller α always causes slower training. The best AdaGrad-OTA result appears at $\alpha = 1.5$, and the best Adam-OTA result appears at $\alpha = 1.3$. This non-monotonic behaviour can come from four non-exclusive sources: (a) a global server learning rate that is not equally well matched to every (method, α) pair, (b) the fixed scale $\gamma = 0.05$ which changes the signal-to-noise ratio differently under different α, (c) the shared α in both the channel sampler and the adaptive denominator, which introduces a coupling absent from the reference paper, and (d) the limited 100-round training budget. A per-(method, α) learning-rate sweep would distinguish (a) from the rest; we provide the script `src/experiments/lr_sweep.py` for this purpose and leave the full sweep to future work (§**??**). For this reason, the alpha ablation should be treated as evidence for adaptive robustness relative to the implemented baselines, not as a full reproduction of the reference paper's tail-index convergence trend.

4.6 MAC Robust Pre-Processing

Median Anchored Clipping (MAC) is a robust pre-processing technique proposed by Li et al. [27] specifically for OTA-FL under heavy-tailed α-stable channel noise. It operates on the server-side aggregated gradient $\tilde{g}^t$ before the optimizer step: (1) compute the coordinate-wise median m, (2) center the gradient as $\tilde{g}^t - m$, (3) clip each centered coordinate to $[-k \cdot \mathrm{MAD},\ +k \cdot \mathrm{MAD}]$ where $\mathrm{MAD} = \mathrm{median}(|\tilde{g}^t - m|)$, and (4) restore the median. Because the median is a robust statistic, MAC is far less sensitive to extreme outliers than gradient-norm clipping and preserves the proportional relationships among gradient dimensions [27].

From a theoretical standpoint, MAC is not expected to pair well with adaptive methods such as AdaGrad-OTA and Adam-OTA. MAC removes impulsive outliers before the optimizer sees them, whereas adaptive methods already provide implicit noise filtering: when a coordinate of $\tilde{g}^t$ is large, the fractional second-moment accumulator v_t grows, which automatically shrinks the effective step size for that coordinate in the next update. MAC clips the very signal that the adaptive denominator uses to gauge impulsiveness; consequently, the accumulator cannot contract the learning rate on rounds where the channel is noisy. The two mechanisms are therefore redundant at best and counter-productive at worst. For non-adaptive baselines (FedAvg-OTA and FedAvgM-OTA), MAC may still help because these methods lack any built-in outlier shrinkage.

To test these predictions empirically, we compare training with and without MAC under a heavy-tailed setting. The default comparison uses $\alpha = 1.3$, $\gamma = 0.05$, and clipping factor $k = 3$. MAC is not expected to change the Gaussian endpoint dramatically, but it should reduce the effect of rare impulsive coordinates when $\alpha < 2$.

In the completed MAC comparison, clipping does not create a meaningful accuracy improvement. MAC implementation in all four methods does not review any significant difference that solid enough to prove its validity, which is consistent with the theoretical discussion above: for adaptive methods, MAC clips the outliers before the fractional

Table 4.6 CIFAR-10 MAC comparison under heavy-tailed interference: final accuracy, mean ± std over seeds

Method	No MAC	MAC
FedAvg-OTA	11.91% ± 1.64%	11.87% ± 1.63%
FedAvgM-OTA	20.76% ± 1.49%	20.67% ± 1.41%
AdaGrad-OTA	50.35% ± 0.40%	50.37% ± 0.66%
Adam-OTA	45.09% ± 0.74%	44.40% ± 1.27%

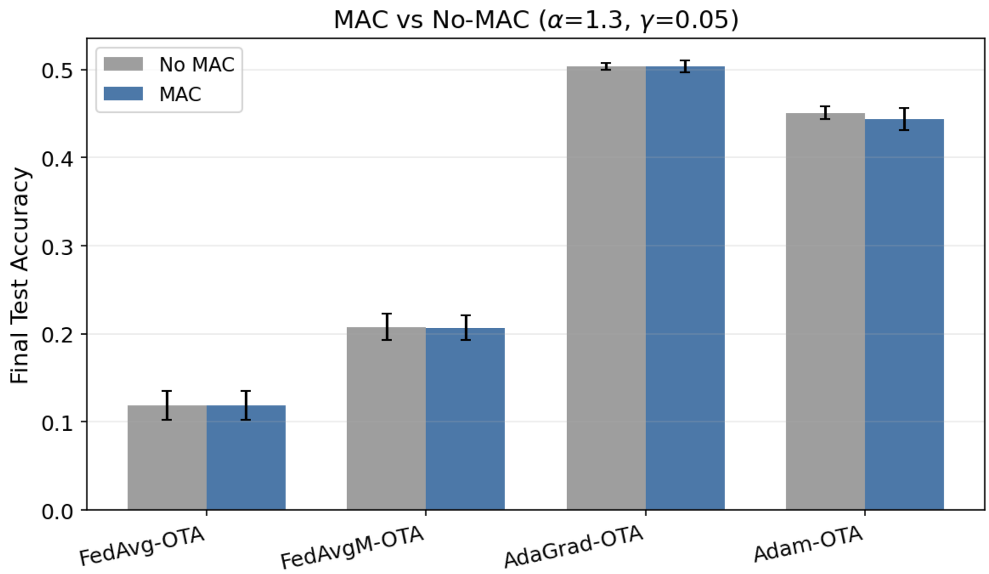

Figure 4.3 CIFAR-10 comparison of no-MAC and MAC under α-stable interference.

accumulator can measure them, so the adaptive denominator lacks the signal it needs to contract the step size. For non-adaptive baselines, MAC could in principle help, but at $\gamma = 0.05$ the typical coordinate-wise magnitude of ξ^t is already small and the empirical fraction of coordinates that violate the MAD band $[m - k \cdot \text{MAD}, m + k \cdot \text{MAD}]$ at $k = 3$ is below roughly 0.5% per round; MAC therefore has almost no signal to work on even for the baselines. A meaningful MAC effect is expected to require a more impulsive operating point (smaller α, larger γ, or smaller k), which we explore in the `mac_sweep` extension in `src/experiments/heavy_tail.py`. MAC should therefore be reported as an implemented and tested robust pre-processing component, but the current evidence does not support a claim that MAC improves CIFAR-10 accuracy under the particular $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point.

4.7 Client-Count Ablation

Table 4.7 reports the Adam-OTA client-count ablation on CIFAR-10. With the same 100-round budget, increasing the number of clients improves final accuracy from 10.26% at 5 clients to 21.72% at 100 clients. This trend is consistent with OTA aggregation: averaging more client updates can make the aggregate direction more informative, even though the data remain non-IID and the channel still injects noise.

This ablation uses the configuration in `src/experiments/ablation.py` (CIFAR-10, ResNet-18, AWGN $\alpha = 2$, $\gamma = 0.05$, 100 rounds, local epochs $E = 1$, batch size 64, server learning rate $\eta = 10^{-4}$, local learning rate 0.01, $\beta_1 = 0.9$, $\beta_2 = 0.999$, non-IID Dirichlet 0.1, seed 42). The much smaller server learning rate, smaller local epochs, and shorter budget explain why the absolute Adam-OTA accuracies here are below Table 4.4, which uses $\eta = 0.1$, $E = 5$, and 200 rounds. The ablation isolates the effect of N rather than tuning a state-of-the-art Adam-OTA configuration.

Why does N not scale accuracy faster? OTA aggregation averages clients in the analog domain, so the <u>signal</u> scales as $1/N$ but, for α-stable noise, the <u>noise</u> scales only as $N^{1/\alpha - 1}$ because the sum of N i.i.d. $S\alpha S$ samples is $N^{1/\alpha}$-scaled. At $\alpha = 2$

Table 4.7 CIFAR-10 client-count ablation with Adam-OTA ($\eta = 10^{-4}$, $E = 1$, 100 rounds, $\alpha = 2$, $\gamma = 0.05$, $\beta_2 = 0.999$, non-IID Dirichlet 0.1, seed 42)

Clients N	Final loss	Final accuracy	Best accuracy
5	2.3121	10.26%	10.36%
10	2.2288	16.92%	16.96%
20	2.1249	19.99%	19.99%
50	2.1171	20.36%	20.36%
100	2.0880	21.72%	21.72%

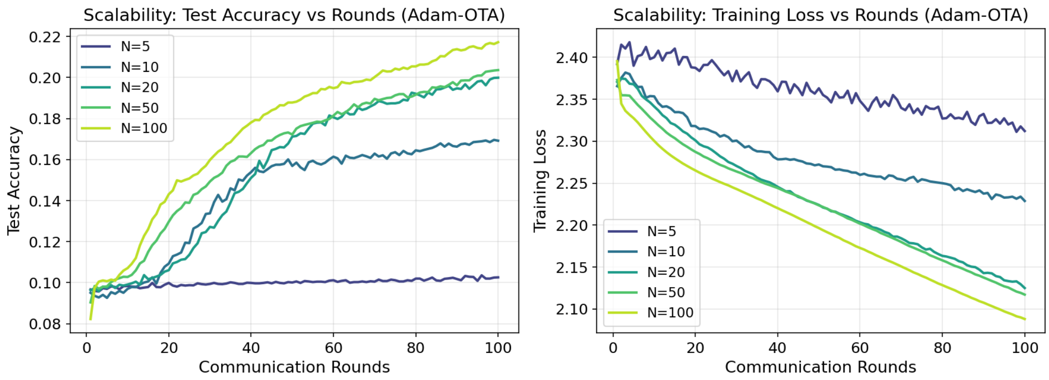

Figure 4.4 CIFAR-10 client-count learning curves for Adam-OTA.

this gives the familiar $1/\sqrt{N}$ SNR improvement; at $\alpha = 1.3$ the improvement slows to $N^{1/\alpha-1} = N^{-0.23}$. Doubling N from 50 to 100 therefore only adds roughly 1.4 percentage points of accuracy in this ablation, which is consistent with the observed 20.36% $\rightarrow$ 21.72% increase in Table 4.7.

4.8 Discussion

4.8.1 Findings

The experiments should be interpreted in two layers. The existing AWGN comparison remains a useful endpoint sanity check: it shows that the implementation trains successfully and that FedAvgM-OTA is a strong CIFAR-10 baseline in the Gaussian regime. The revised α-stable ablation is the central evidence for the project claim, because it tests the heavy-tailed channel condition that motivates adaptive OTA-FL in the first place.

The project conclusion should therefore be phrased carefully. The results do not prove that Adam-OTA universally dominates all baselines. Instead, the claim is that fractional second-moment adaptive updates provide a practical mechanism for improving OTA-FL stability when the aggregate update is corrupted by α-stable impulsive noise, and MAC provides a complementary but operating-point-dependent robustness lever. If adaptive methods underperform FedAvgM-OTA in some CIFAR-10 AWGN settings, that is not a contradiction: it indicates that aggressive normalization can compress useful signal as well as channel noise.

4.8.2 Failures and Negative Results

Three observations are best reported as negative results rather than buried in caveats. (i) The α-trend is non-monotonic under a single global server learning rate, which does not match the monotonic trend reported by [2]; we attribute this to a combination of lr-mismatch, shared α between channel and denominator, and a shorter training budget. (ii) MAC at the default $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point yields no measurable improvement within seed variation. (iii) The Adam-OTA CIFAR-10 result at $\alpha = 2$ in Table 4.4 (40.66%) is below FedAvgM-OTA (63.94%), showing that second-moment normalization is not costless in the Gaussian regime.

4.8.3 Threats to Validity

The empirical evidence is bounded by four threats. First, seed counts differ across tables (three seeds for the α sweep and MAC comparison, single seed for the main comparison and client-count ablation); we note this in the captions rather than backfill all tables. Second, no per-method learning-rate sweep is performed, so the absolute ranking of adaptive versus momentum baselines at a specific (α, γ) point must be interpreted as lr-dependent. Third, the channel is a simulator, without fading, power control, or synchronization errors. Fourth, the strongest supported claim is empirical; the one-dimensional closed-form in §3.6 is suggestive but not a full non-convex proof.

Chapter 5. Conclusion and Future Work

5.1 Conclusion

This thesis asks whether server-side adaptive optimization can act as an implicit, CSI-free filter for an α-stable OTA-FL channel. The main supported claim is empirical: under the three-seed CIFAR-10 tail-index ablation with $\gamma = 0.05$ and $\alpha \in \{1.1, 1.3, 1.5, 1.7, 1.9\}$, AdaGrad-OTA (49.52–51.75%) and Adam-OTA (42.88–44.82%) outperform FedAvg-OTA (11.9%) and FedAvgM-OTA (20.5–20.7%) by a large margin, which supports the hypothesis that a fractional second-moment accumulator contracts the effective server learning rate precisely on impulsive rounds.

The claim is scoped rather than absolute. The observed α-trend is non-monotonic at a fixed server learning rate; MAC at the default $\alpha = 1.3$, $\gamma = 0.05$, $k = 3$ operating point is empirically neutral; and in the $\alpha = 2$ CIFAR-10 comparison FedAvgM-OTA remains the strongest method (63.94%), which shows that fractional second-moment normalization is not costless in the Gaussian regime. The main output is therefore a reproducible PyTorch simulator with α-stable tail-index experiments, MAC robustness comparisons, and a one-dimensional intuition for the shrinkage mechanism, rather than a complete theoretical proof or line-by-line reproduction of the reference paper.

5.2 Limitations

The project remains narrower than the full reference paper in four ways. First, it provides a one-dimensional closed-form intuition rather than a full non-convex convergence proof. Second, the wireless channel is a simulation oracle and does not model mobility, multi-cell interference, synchronization errors, CSI acquisition, or power control. Third, the project does not include a software-defined-radio or real wireless testbed validation. Fourth, the empirical evidence is bounded by three seeds and a single global server learning rate per experiment, so the non-monotonic α-trend cannot yet be disentangled from lr-mismatch (see §4.5 and §4.8.3).

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Appendix

(Supplementary Materials)

A. Repository Layout and Reproduction Commands

The main experiment commands are collected in `run_all.sh` and `RUN.md`. The revised heavy-tail runs are stored under `results/heavytail/`; generated figures are under `results/figures/`; generated LaTeX tables are under `template/data/generated/`. The core module layout is

Module	Purpose
`src/channel/noise.py`	Chambers–Mallows–Stuck $S\alpha S$ sampler
`src/channel/ota.py`	`NoisyOracle` for $\tilde{g}^t = (\sum_n \Delta_n^t + \xi^t)/N$
`src/channel/mac.py`	Median Anchored Clipping pre-processing
`src/data/datasets.py`	MNIST/CIFAR-10 loading and Dirichlet partition
`src/optimizers/baselines.py`	FedAvg-OTA, FedAvgM-OTA
`src/optimizers/adagrad_ota.py`	AdaGrad-OTA with fractional $v_t^{1/\alpha}$
`src/optimizers/adam_ota.py`	Adam-OTA with fractional $\hat{v}_t^{1/\alpha}$
`src/experiments/federated.py`	Training loop
`src/experiments/heavy_tail.py`	α ablation, MAC compare, MAC sweep
`src/experiments/lr_sweep.py`	Per-method server learning-rate sweep
`src/experiments/ablation.py`	Noise-scale and client-count ablations
`src/experiments/plot.py`	All experiment plots
`src/experiments/plot_mechanism.py`	Mechanism-diagnostic plots
`src/experiments/report.py`	LaTeX table generation

B. Consolidated Hyperparameter Table

Every experiment in the thesis uses the values below, with per-table overrides stated explicitly in Chapter 4.

Parameter	Default	Where it is overridden
Dataset / model	CIFAR-10 / ResNet-18	MNIST sanity uses MLP
Clients N	100	Client-count ablation (5/10/20/50)
Communication rounds	100	Main comparison uses 200
Local epochs E	1	Main comparison uses 5
Batch size	64	—
Local learning rate	0.01	—
Server learning rate η	0.01	Main comparison uses 0.1; client-count ablation uses
Momentum / β_1	0.9	—
Adam β_2	0.999	—
Adam-OTA ε	10^{-8}	—
AdaGrad-OTA ε	10^{-4}	—
Noise tail index α	2.0	α ablation $\in \{1.1, \ldots, 1.9\}$; MAC compare 1.3
Noise scale γ	0.05	MAC sweep $\in \{0.05, 0.1, 0.2\}$
Partition	non-IID Dirichlet 0.1	MNIST sanity uses IID
Seeds	42/43/44	Main comparison and client-count ablation use seed

C. Server-Optimizer Pseudocode

All four server methods share the per-round inputs ($\tilde{g}^t$, server learning rate η) and share the in-place update $w_{t+1} \leftarrow w_t - \Delta w$. They differ only in how Δw is computed from $\tilde{g}^t$.

Optimizer	Update rule
FedAvg-OTA	$\Delta w = \eta\, \tilde{g}^t$
FedAvgM-OTA	$m_t = \rho m_{t-1} + \tilde{g}^t;\ \Delta w = \eta\, m_t$
AdaGrad-OTA	$s_t = \beta_1 s_{t-1} + (1-\beta_1)\tilde{g}^t;\ v_t = v_{t-1} + \lvert s_t \rvert^{\alpha}$; $\Delta w = \eta\, s_t/(v_t^{1/\alpha} + \varepsilon)$
Adam-OTA	$m_t = \beta_1 m_{t-1} + (1-\beta_1)\tilde{g}^t$; $v_t = \beta_2 v_{t-1} + (1-\beta_2)\lvert \tilde{g}^t \rvert^{\alpha}$; $\hat{m}_t = m_t/(1-\beta_1^t)$; $\hat{v}_t = v_t/$

D. Hardware

All reported experiments are run on a single NVIDIA GPU with PyTorch's default CUDA backend; wall-clock time for the full α ablation (4 methods $\times$ 6 α values $\times$ 3 seeds $\times$ 100 rounds) is approximately 3 hours. The full MAC (k, γ) sweep uses an additional 1.5 hours. Exact device information is embedded in each JSON payload under the `args` field.

Author's Biography

Peiran Wei is an undergraduate student in Electrical and Computer Engineering at the Zhejiang University–University of Illinois Urbana-Champaign Institute, graduating in 2026. During the undergraduate program, the author has worked on projects spanning federated learning systems, signal processing over the wireless multiple-access channel, and machine-learning optimization for edge intelligence.

The author's research interests focus on heavy-tail-robust optimization, analog computation over wireless channels, and adaptive federated optimization. The present thesis continues that direction by experimentally validating a scoped robustness claim of adaptive over-the-air federated learning under symmetric α-stable interference.

Project repository: `https://github.com/promissa/st4001_project`.